

Documentation for nmth.chpl

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1 logmod: log-modulus function

1.1 Use

```
inline proc logmod(  
    const in x: real  
): real {
```

1.2 Theory

The *log-modulus* function was introduced by [John and Draper \(1980\)](#). It is asymptotically equal to x for small values of its argument, and asymptotically equal to $\ln(\cdot)$ for large positive values of its argument:

$$\text{logmod}(x) \equiv \text{sign}(x) \ln(|x| + 1). \quad (1)$$

Note that, asymptotically,

$$\text{logmod}(x) \sim \begin{cases} x, & x \sim 0, \\ \text{sign}(x) \ln(|x|), & |x| \gg 1, \end{cases} \quad (2)$$

and that $\text{logmod}(x)$ is odd.

2 expmod: exponential-modulus function

2.1 Use

```
inline proc expmod(  
    const in x: real  
): real {
```

2.2 Theory

We often need the inverse of logmod:

$$\begin{aligned}y &= \text{sign}(x) \ln(|x| + 1), \\ \text{sign}(y)y &= \ln(|x| + 1),\end{aligned}$$

(because logmod is odd)

$$\begin{aligned}\exp(\text{sign}(y)y) &= |x| + 1, \\ \exp(|y|) &= |x| + 1, \\ |x| &= \exp(|y|) - 1, \\ x &= \text{sign}(y) [\exp(|y|) - 1],\end{aligned}$$

whence

$$\text{expmod}(x) \equiv \text{sign}(x) [\exp(|x|) - 1] \tag{3}$$

is the *exponential-modulus* function.

References

John, J. and Draper, N. R. (1980). An alternative family of transformations. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 29(2):190–197.